Gijs Bartholomeus

Curriculum Vitae

21 October 2026

Oxford, United Kingdom | linkedin.com/in/gijsbartholomeus

Education

DPhil in Mathematics

Oct 2026 – Jul 2030

University of Oxford

Research in mathematical biology, algebraic topology, and topological data analysis. Supervisors: Helen Byrne, Alain Goriely, Lida Kanari, and Heather Harrington.

Double MSc in Physics and Mathematics, cum laude

Sep 2023 – Jul 2026

Utrecht University

Weighted average: 8.7/10[*].
Thesis: *Simplicity Bias and Sloppiness in Gene Regulatory Networks*, completed as a Recognised Student at the University of Oxford under the supervision of Ard Louis.

Double BSc in Physics and Mathematics, cum laude

Sep 2020 – Jul 2022

Utrecht University

Thesis: *Stochastic extinction dynamics in Lotka–Volterra models using Random Matrix Theory.*

MSc in Medicine

Sep 2022 – Jul 2024

Utrecht University

Clinical management courses and full-time hospital residencies. Research project on variational autoencoders for harmonising medical imaging data.

BSc in Medicine, cum laude

Sep 2017 – Jul 2021

Utrecht University

[*]The Dutch grading scale runs from 1 to 10; 6 is a pass, and grades of 9 or 10 are rarely awarded. Cum laude requires a weighted average of at least 8.0.

Research Experience

Junior Researcher

Dec 2018 – Sep 2022

Department of Radiology, Utrecht University

Investigated the reproducibility of quantitative CT image analysis (radiomics), contributing to work on the robustness of pulmonary-nodule features across radiation dose levels.

MSc Research, Recognised Student

2025 – 2026

University of Oxford

Using ODE models of the budding-yeast cell cycle, identified a frequency–complexity envelope in which highly complex dynamical phenotypes were not frequent, while wild-type-like oscillations occupied a relatively high-frequency region. Developed a nested neutral-set estimator for rare phenotype-preserving volumes and related these global biases to locally sloppy parameter-space geometry.

Publications

1. **Bartholomeus, G. A.**, van Amsterdam, W. A. C., den Harder, A. M., Willemink, M. J., van Hamersvelt, R. W., de Jong, P. A., and Leiner, T. (2023). Robustness of pulmonary nodule radiomic features on computed tomography as a function of varying radiation dose levels: a multi-dose in vivo patient study. *European Radiology*, 33, 7044–7055.

2. van Amsterdam, W. A. C., Verhoeff, J. J. C., Harlianto, N. I., **Bartholomeus, G. A.**, Puli, A. M., de Jong, P. A., Leiner, T., van Lindert, A. S. R., Eijkemans, M. J. C., and Ranganath, R. (2022). Individual treatment effect estimation in the presence of unobserved confounding using proxies: a cohort study in stage III non-small cell lung cancer. *Scientific Reports*, 12, 5848.

Teaching Experience

Teaching Assistant

Sep 2020 – Jul 2026

Utrecht University

Organised and taught courses across the departments of physics, mathematics, and medicine.

Senior Teacher

Jan 2018 – Jul 2025

Stichting Studiebegeleiding, Leiden

Delivered intensive one- to four-day examination-preparation courses for classes of up to 30 secondary-school students and led a team of seven teaching assistants.

Professional Experience

Medical Technology Specialist

Nov 2025 – present

Medscio Interoperability B.V.

Map clinical laboratory parameters from heterogeneous hospital systems to standard terminologies. Built and benchmarked LLM-based extraction pipelines and developed a SapBERT embedding search engine over SNOMED CT for Dutch cardiology data.

Science Trainee

Jan 2023 – May 2023

MIcompany (now Rewire)

Built a LightGBM subscription-pricing model for a national golf federation, supporting recommendations on membership tiers and price points.

Leadership and Activities

Committee Chair, Triton Career Days

2021 – 2022

U.S.R. "Triton"

Chaired the committee responsible for the student rowing association's annual careers event.

Competitive Rower

Sep 2017 – May 2022

U.S.R. "Triton"

Competed at national level while training up to 20 hours per week.

Technical Skills

Programming	Python (NumPy, SciPy, PyTorch, TensorFlow), R, MATLAB, C#, Lean/mathlib (basic familiarity), Git, Linux
Mathematics	Algebraic topology, topological data analysis, dynamical systems, differential and information geometry, random matrix theory, probability
Methods	ODE modelling, Monte Carlo methods, high-dimensional sampling, optimisation, dimensionality reduction, machine learning